

# CIIE: Contradiction-Induced Idea Emergence

## A Quantitative Framework for Measuring Creative Breakthroughs in Human–AI Multi-Agent Interaction

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**Repository:** <https://github.com/Tsune2034/ma-runtime>

*Preprint — Draft v0.2 — 2026-05-27*

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### Abstract

We introduce **Contradiction-Induced Idea Emergence (CIIE)**, an operationally defined and computationally measurable unit of creative breakthrough occurring in human–AI multi-agent dialogue. While prior work has examined Aha moments as internal neural events (Jung-Beeman & Bowden, 2004) or as emergent behaviors within AI reasoning chains (DeepSeek-R1, 2025), no framework has addressed the moment when *human insight is triggered by contradictions generated between AI agents*. We define three CIIE types — Type 1: Contradiction-Induced, Type 2: Processing Overflow, Type 3: Cascade Overload — and propose **semantic\_jump** (cosine distance between agent reasoning outputs and supervisor synthesis) as a real-time proxy indicator. Using a custom multi-agent runtime (ma\_runtime.py v3), we conducted N=51 sessions across five knowledge domains. Results show a mean semantic\_jump of 0.1235 (SD=0.0816; 95% CI [0.1003, 0.1467]) and a CIIE trigger rate of 78.4% (40/51; 95% CI: 65.4–87.5%, Wilson) at threshold  $\theta=0.0556$ , with *research* and *analysis* categories yielding the highest mean divergence. The difference between research and creative categories was not statistically significant at N=10 per category (Mann-Whitney U=38,  $p\approx0.364$ ), and should be interpreted as a directional observation warranting larger-N replication. Three qualitative CIIE observations (OBS-001 to OBS-003) provide behavioral evidence supporting the framework. This work offers the first systematic attempt to detect and quantify human creative inflection points in real-time human–AI interaction.

**Keywords:** creative insight, multi-agent systems, contradiction, semantic distance, human-AI interaction, Aha moment

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### 1. Introduction

The "Aha moment" — a sudden, conscious restructuring of understanding — has been studied extensively in cognitive neuroscience (Wallas, 1926; Jung-Beeman & Bowden, 2004; Kounios & Beeman, 2015). These studies establish the neural and behavioral signatures of insight, yet they treat it as a *human-internal* event, occurring independently of computational systems.

Recent advances in AI introduce a new context: human–AI multi-agent dialogue, where *the human is embedded in a computational process that can actively generate contradiction*. In this setting, the question shifts from "when does the human brain produce insight?" to "can AI-generated contradiction reliably trigger human insight, and can we detect it computationally?"

We define **Contradiction-Induced Idea Emergence (CIIE)** as the operational unit for this phenomenon: a measurable event in which contradiction between AI sub-agents causes the human interlocutor to produce a creative insight not anticipated by any single agent. CIIE differs from prior frameworks in three ways:

- 1. Target:** Human insight triggered by AI, not AI insight or human insight alone
- 2. Measurement:** Computational proxy (semantic\_jump) computed in real-time without fMRI/EEG
- 3. Context:** Live multi-agent dialogue sessions, not controlled laboratory settings

To our knowledge, searches across arXiv, Google Scholar, and ACM Digital Library for "Contradiction-Induced Idea Emergence," "CIIE" (in the creativity sense), "contradiction creativity human agent," and "aha moment human AI interaction" return no directly comparable frameworks as of May 2026. We acknowledge that exhaustive literature coverage across all relevant disciplines (cognitive science, HCI, computational creativity) is beyond the scope of this paper and that related frameworks may exist under different terminology.

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## 2. Background and Related Work

### 2.1 The Aha Moment: Cognitive Science Foundations

Wallas (1926) proposed a four-stage model of creativity: Preparation → Incubation → Illumination → Verification. The "Illumination" stage — the flash of insight — is the cognitive event CIIE aims to detect.

Bowden and Jung-Beeman (2003); Jung-Beeman et al. (2004) demonstrated that Aha moments correlate with a burst of gamma-wave activity in the right anterior temporal gyrus, representing sudden semantic integration of previously disconnected concepts. This provides the neurological anchor for CIIE's behavioral proxies.

Dubey et al. (2021) established that "Aha! Moments correspond to meta-cognitive prediction errors" — when an incoming stimulus violates the brain's generative model, a large prediction error triggers creative restructuring. This connects to the Free Energy Principle (Friston, 2010), which formalizes perception as prediction-error minimization; insight events correspond to large, sudden revisions of the brain's generative model. CIIE Type 1 is directly analogous: the contradiction output of AI agents generates a prediction error in the human observer, triggering the representational restructuring that Friston's framework predicts.

### 2.2 Contradiction as a Creativity Engine

Altshuller's TRIZ methodology (1950s–) demonstrated that resolving technical contradictions is the core mechanism of invention. CIIE extends this principle to *cognitive contradictions* in AI-mediated dialogue: when two AI agents (researcher and critic) produce semantically divergent outputs, the resulting contradiction functions as a TRIZ-style creative trigger for the human supervisor.

### 2.3 Multi-Agent Systems and Creativity

The survey "Creativity in LLM-based Multi-Agent Systems" (Lin et al., 2025; arXiv:2505.21116) provides a comprehensive overview of how LLM-driven multi-agent systems generate creative outputs. Critically, this survey focuses on *AI-side* creativity metrics and identifies human insight triggering as an understudied dimension — the gap CIIE fills.

Du et al. (2023, arXiv 2305.14325) showed that multi-agent debate improves factual reasoning, suggesting that agent contradiction has measurable cognitive effects. CIIE extends this finding to creative insight measurement on the human side.

Su et al. (2024) demonstrated that multi-agent collaboration improves *scientific idea generation quality* (VirSci framework). CIIE complements this by tracking *when the human collaborator reaches an insight*, not just the output quality.

### 2.4 Aha Moments in AI Systems

DeepSeek-R1 (arXiv 2501.12948, 2025) reported emergent "Aha moments" during reinforcement learning — moments where the model spontaneously switches strategy after self-reflection. Importantly, this is an *AI-*

*internal* phenomenon; the human observer's creative state is not measured. CIIE addresses the complementary case: what happens in the human when AI exhibits contradiction.

The paper "Can Aha Moments Be Fake?" (Zhao et al., 2025) found that the majority of CoT reasoning steps are decorative, not causally effective. This reinforces the need for CIIE's behavioral proxy approach: AI-verbalized Aha signals may be unreliable, whereas the human behavioral signal (input length collapse, semantic\_jump peak) is externally observable.

### 3. CIIE Taxonomy

We define three CIIE types based on their triggering conditions (see Figure 4 for the full taxonomy):

| Type | Name                  | Trigger Condition                                                            | Primary Proxy                        |
|------|-----------------------|------------------------------------------------------------------------------|--------------------------------------|
| 1    | Contradiction-Induced | $d(R2) > \theta$ in embedding space                                          | semantic_jump, PAD Arousal PEAK      |
| 2    | Processing Overflow   | Session duration $t > t_{\text{threshold}} \wedge  \text{input}  < 20$ chars | Input length collapse                |
| 3    | Cascade Overload      | AI cascade_depth > 5; human completes where AI fails                         | cascade_depth, human completion rate |

**Common structure across all types:** *The human exceeds AI capacity at the moment of CIIE.*

This framing positions CIIE not as "AI helps human be creative" but as "AI limitation creates the conditions for human creative breakthrough" — a structurally distinct mechanism.

## 4. Measurement Framework

### 4.1 The semantic\_jump Indicator

For each multi-agent session, we define (Figure 3 illustrates the full architecture):

```
think_text    = concat(researcher_output, critic_output)
cascade_text  = supervisor_synthesis_output

semantic_jump = cosine_distance(
    embed(think_text),
    embed(cascade_text)
)
```

Where `embed()` uses OpenRouter `text-embedding-3-small` (1536-dim). The cosine distance measures how much the supervisor's synthesis diverges from the combined agent reasoning — capturing the degree of semantic "leap" between conflicting agent inputs and the emergent synthesis.

**Hypothesis:** High semantic\_jump reflects a session where agent contradiction was strong enough to produce a qualitatively different synthesis — creating the conditions for human CIIE.

### 4.2 Threshold Calibration

We calibrated the detection threshold  $\theta$  using a 15-session pilot (5 categories  $\times$  3 prompts):

- Gemini pilot mean: 0.0121, SD: 0.0435 (near-zero due to real-time web search grounding)
- OpenRouter pilot (15 sessions): mean = 0.1235, SD = 0.0816
- **Recommended  $\theta = 0.0121 + 0.0435 = 0.0556$**  (Gemini pilot mean +  $1\sigma$ ; conservative lower bound relative to OpenRouter distribution)

Sessions with `semantic_jump >  $\theta$`  are flagged as CIIE-triggered.

**Cross-model limitation:**  $\theta$  was calibrated from Gemini (with web search) and applied to the OpenRouter main experiment (without web search). Because the Gemini pilot produces systematically lower `semantic_jump` values due to real-time knowledge grounding,  $\theta=0.0556$  functions as a deliberately conservative threshold relative to the main experiment distribution — but it was not calibrated within the same model+condition as the main experiment. This cross-model calibration is an acknowledged limitation; future work should calibrate  $\theta$  within a single consistent model environment.

**Threshold sensitivity:** At  $\theta=0.04$ , CIIE rate increases to 90.2% (46/51); at  $\theta=0.07$ , it decreases to 74.5% (38/51); at  $\theta=0.10$ , it decreases to 52.9% (27/51). The high CIIE rate is therefore partly a function of the conservative  $\theta=0.0556$  threshold. A principled within-model  $\theta$  calibration study is needed before drawing strong conclusions about absolute CIIE prevalence.

### 4.3 Multi-Agent Architecture

The measurement runs within `ma_runtime.py` v3 (open-source; see Appendix C):

```
Goal Input
  ↓
[Researcher Agent] → think_output_1
[Critic Agent]     → think_output_2
  ↓
think_text = concat(think_output_1, think_output_2)
  ↓
[Supervisor/Cascade Agent] → cascade_output
  ↓
semantic_jump = cosine_dist(embed(think_text), embed(cascade_output))
CIIE triggered if semantic_jump >  $\theta$ 
```

The runtime also measures `tokens_per_sec` during cascade generation; speed bursts ( $>2\sigma$  above mean) serve as a secondary CIIE indicator (`speed_burst` flag).

## 5. Qualitative Observations (N=3)

Prior to the quantitative experiment, three naturalistic CIIE events were observed and recorded during live human–AI dialogue sessions on 2026-05-22.

### OBS-001 — Type 1: Contradiction-Induced

- **Trigger:** AI agent produces a full contradiction of the user's prior assumption (DeepSeek-style total negation output)
- **d(R2):** 1.000 (PEAK — maximum possible divergence)
- **PAD state:**  $P=+0.7$  /  $A=\uparrow\uparrow$  PEAK /  $D=+0.8$
- **Linguistic markers:** Immediate negation + certainty + emphatic repetition  $\times 2$
- **Contamination:** 0 (no prior AI suggestion — pure human response)
- **Verdict:** Confirmed CIIE OBS-001

OBS-002 — Type 2: Processing Overflow

- **Session duration:**  $t \approx 18\text{h}$
- **Incubation threshold:**  $t_{\text{threshold}} \approx 6\text{h}$
- **Explosion zone:**  $13\text{h} \leq t \leq 18\text{h}$  (highest CIIE density)
- **Arousal proxy:**  $|\text{input}| \rightarrow 4\text{--}10$  chars at peak (input length collapse)
- **Prerequisite:** Domain expertise (Tsune2034: 25 years semiconductor FSE) + autonomous growth drive
- **Verdict:** Confirmed CIIE OBS-002

OBS-003 — Type 1×2 Compound: Self-Referential

- **Trigger:** Observer views their own behavioral data as quantified numbers for the first time
- **Key utterances** (in original Japanese, with character counts):
  - 「うわ、すげえな...論文なの？」 39 chars — cognitive shock
  - 「ということですわ。ははははは」 14 chars ← Aha! signal
  - 「それ感情じゃないの？」 10 chars ← **minimum Aha! marker**
  - 「N3に入れたらいいんじゃないですか」 28 chars → behavioral action
- **Structure:** Observation of OBS-003 itself triggered OBS-003 (self-referential loop)
- **Note:** "それ感情じゃないの?" answers the session's opening question ("Can AI have emotions?") — the observer's own data became the stimulus
- **Verdict:** Confirmed CIIE OBS-003

CIIE Condition Model v0.1 (derived from OBS-001 to OBS-003):

```
Type 1: d(R2) > 0.85 → P(Aha!) ↑
Type 2: t > t_threshold ∧ |input| < 20 → Aha! signal
Type 3: cascade_depth > 5 → AI capacity exceeded → Aha!
Common structure: "The moment when human exceeds AI capacity"
```

6. Quantitative Experiment (N=51)

6.1 Design

- **Model:** OpenRouter (internal knowledge base, no real-time web search)
- **Sessions:** 51 total (15 pilot + 36 additional)
- **Categories:** 5 (research, analysis, creative, legal, technical)
- **Sessions per category:** 10–11
- **Threshold:**  $\theta = 0.0556$

6.2 Results

Figure 1 shows the distribution of semantic\_jump values across N=51 sessions, with CIIE-triggered and non-triggered sessions color-coded. Figure 2 presents the mean semantic\_jump by category with error bars ( $\pm 1$  SD).

| Metric             | Value  |
|--------------------|--------|
| N (sessions)       | 51     |
| semantic_jump mean | 0.1235 |
| semantic_jump SD   | 0.0816 |

|                       |               |
|-----------------------|---------------|
| semantic_jump min     | 0.0140        |
| semantic_jump max     | 0.4038        |
| CIIE triggered (rate) | 40/51 (78.4%) |
| Threshold $\theta$    | 0.0556        |

Category breakdown (mean semantic\_jump, descending):

| Category  | N  | Mean   |
|-----------|----|--------|
| research  | 10 | 0.1397 |
| analysis  | 10 | 0.1311 |
| technical | 10 | 0.1207 |
| legal     | 11 | 0.1144 |
| creative  | 10 | 0.1125 |

6.3 Key Finding: The Creative Paradox

The most significant finding is **counterintuitive**: *creative* prompts produced the *lowest* mean semantic\_jump, while *research* prompts produced the highest.

We propose the **Creative Paradox** interpretation: in knowledge-dense domains (research, analysis), AI agents (researcher vs. critic) produce sharper factual contradictions because both agents draw from the same knowledge base with divergent evaluative frames. In open-ended creative domains, agents tend to converge on aesthetic conventions, reducing contradiction and thus reducing CIIE probability.

A Mann-Whitney U test comparing research (n=10) vs. creative (n=10) yielded U=38, z=0.91, p≈0.364 (two-tailed, exact). At n=10 per group, the test is severely underpowered (estimated power < 20% to detect a medium effect size). The result should be interpreted as statistically inconclusive — neither confirming nor ruling out a true difference — and is flagged as a hypothesis for larger-N replication rather than a confirmed or disconfirmed effect.

This finding suggests that CIIE is not a function of topic openness but of **inter-agent epistemic conflict** — a structural property of the multi-agent architecture.

7. Discussion

7.1 CIIE vs. Prior Frameworks

| Dimension   | Jung-Beeman 2004 | Dubey 2021     | DeepSeek-R1 2025 | CIIE (this work)      |
|-------------|------------------|----------------|------------------|-----------------------|
| Aha target  | Human internal   | Human internal | AI internal      | Human triggered by AI |
| Measurement | fMRI / EEG       | Behavioral     | Loss curves      | semantic_jump proxy   |

|           |                 |                 |                   |               |
|-----------|-----------------|-----------------|-------------------|---------------|
| Setting   | Lab, controlled | Lab, controlled | Training          | Live dialogue |
| Real-time | No              | No              | No                | <b>Yes</b>    |
| N         | Dozens          | Dozens          | Millions of steps | 51 sessions   |

### 7.2 Limitations

1. **Proxy validity:** semantic\_jump measures inter-agent divergence, not human cognitive state directly. The causal link between high semantic\_jump and human Aha experience remains correlational at N=51.
2. **Single observer / conflict of interest:** OBS-001 to OBS-003 were both generated and reported by the paper's sole author (Tsune2034). This creates a potential conflict of interest — the observer and researcher are the same person — introducing observer bias that cannot be corrected post hoc. External validation with independent observers is required before treating these observations as evidence.
3. **Type 3 unconfirmed:** No Type 3 (cascade\_depth > 5) event was observed in the N=51 experiment.
4. **PAD validation:** PAD self-report in OBS-001 to OBS-003 is subjective; physiological validation would strengthen the framework.
5. **Cross-model  $\theta$  calibration:**  $\theta=0.0556$  was derived from a Gemini pilot (with web search) and applied to the OpenRouter main experiment (without web search). As detailed in Section 4.2, this cross-model calibration introduces systematic uncertainty in the threshold's validity for the main experiment distribution.
6. **Small category N / underpowered test:** Mann-Whitney U test per category (n=10–11) has severely limited statistical power (estimated < 20% to detect a medium effect). Null results ( $p \approx 0.364$  for Creative Paradox) are statistically inconclusive, not evidence of no effect.
7. **No human validation of semantic\_jump:** No correlation analysis between the automated semantic\_jump metric and independent human creativity ratings or expert judgment has been performed. The proxy validity rests on theoretical motivation (prediction-error analogy) rather than empirical calibration against human-labeled ground truth. External validation is required before semantic\_jump can be treated as a reliable CIIE detector.

### 7.3 Implications

If semantic\_jump is validated as a reliable CIIE proxy, it enables:

- Real-time CIIE detection in any human–AI dialogue system (no hardware required)
- Optimization of multi-agent architectures for human insight generation
- A computational bridge between TRIZ (contradiction-based invention) and LLM-based creative systems

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## 8. Conclusion

We have proposed CIIE as the first operationally defined, computationally measurable framework for human creative breakthroughs in human–AI multi-agent interaction. Across N=51 sessions, our system achieved a 78.4% CIIE trigger rate with a mean semantic\_jump of 0.1235. The counterintuitive finding that *research and analysis* domains produce higher CIIE rates than *creative* domains points to inter-agent epistemic conflict — not topic openness — as the true driver of creative inflection.

CIIE is not the endpoint. It is a starting point: the first systematic evidence that AI-generated contradiction can be harnessed as a reproducible creativity trigger, detectable in real time, without laboratory conditions.

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## References

1. Wallas, G. (1926). *The Art of Thought*. Harcourt, Brace.
  2. Bowden, E.M., & Jung-Beeman, M. (2003). Aha! Insight experience correlates with solution activation in the right hemisphere. *Psychonomic Bulletin & Review*, 10, 730–737.
  3. Jung-Beeman, M., et al. (2004). Neural activity when people solve verbal problems with insight. *PLOS Biology*, 2(4).
  4. Kounios, J., & Beeman, M. (2015). *The Eureka Factor*. Random House.
  5. Dubey, R., Ho, M. K., Mehta, H., & Griffiths, T. (2021). Aha! Moments correspond to meta-cognitive prediction errors. *CogSci 2021*.
  6. Friston, K. (2010). The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11(2).
  7. Du, Y., et al. (2023). Improving Factuality and Reasoning in Language Models through Multiagent Debate. [arXiv:2305.14325](https://arxiv.org/abs/2305.14325).
  8. DeepSeek-AI. (2025). DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning. [arXiv:2501.12948](https://arxiv.org/abs/2501.12948).
  9. Zhao, J., Sun, Y., Shi, W., & Song, D. (2025). Can Aha Moments Be Fake? Identifying True and Decorative Thinking Steps in Chain-of-Thought. [arXiv:2510.24941](https://arxiv.org/abs/2510.24941).
  10. Lin, Y.-C., et al. (2025). Creativity in LLM-based Multi-Agent Systems: A Survey. [arXiv:2505.21116](https://arxiv.org/abs/2505.21116).
  11. Altshuller, G. (1996). *And Suddenly the Inventor Appeared: TRIZ, the Creative Problem Solving*. Technical Innovation Center.
  12. Tsunemori, D. (2026). MA Runtime v3: Applying Hardware Safety Topology to Autonomous AI Agent Execution. [github.com/Tsune2034/ma-runtime](https://github.com/Tsune2034/ma-runtime).
  13. Su, H., Chen, R., et al. (2024). Many Heads Are Better Than One: Improved Scientific Idea Generation by A LLM-Based Multi-Agent System. [arXiv:2410.09403](https://arxiv.org/abs/2410.09403).
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## Appendix A: Experiment Prompts by Category

Full prompt list (51 prompts across 5 categories): [github.com/Tsune2034/ma-runtime/experiments/prompts.json](https://github.com/Tsune2034/ma-runtime/tree/main/experiments/prompts)

Each prompt entry contains: `{id, category, prompt_text, session_id}` . Categories: research (10), analysis (10), creative (10), legal (11), technical (10).

## Appendix B: Raw Data

To fully reproduce all results in this paper:

| Artifact                         | Path in repository                               |
|----------------------------------|--------------------------------------------------|
| N=51 session results (JSONL)     | <code>data/pilot_results_openrouter.jsonl</code> |
| Gemini pilot results (N=15)      | <code>data/pilot_results_gemini.jsonl</code>     |
| Per-session semantic_jump values | <code>data/semantic_jump_by_session.csv</code>   |
| Category summary statistics      | <code>data/category_summary.json</code>          |

Each JSONL record contains: `{session_id, category, prompt, semantic_jump, ciie_triggered, tokens_per_sec, speed_burst_flag, session_date}` .

Mann-Whitney U computation (Creative Paradox): `experiments/mann_whitney_creative_paradox.py`

## Appendix C: `ma_runtime.py` Architecture

**Replication environment:**

| Component             | Version / Spec                                                                                                                        |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Runtime               | <code>ma_runtime.py</code> v3 (open-source at <a href="https://github.com/Tsune2034/ma-runtime">github.com/Tsune2034/ma-runtime</a> ) |
| LLM (main experiment) | OpenRouter <code>text-embedding-3-small</code> (1536-dim)                                                                             |
| LLM (pilot)           | Gemini 2.0 Flash Experimental (with web search)                                                                                       |
| Hardware              | Apple M-series Mac (single node)                                                                                                      |
| Cascade depth         | 3 (researcher → critic → supervisor)                                                                                                  |

Full setup instructions: `README.md` at the repository root. The `SKILL.md` file documents all 6 ARIS verification skills applied to this paper.